

STATEMENT OF KEVIN KAWCHAK , CEO CHEMICALQDEVICE
BEFORE THE COMMITTEE ON ENERGY AND COMMERCE, SUBCOMMITTEE ON HEALTH
U. S. HOUSE OF REPRESENTATIVES

Verifying Surgical-Robot Software Before It Touches a Patient

10.5281/zenodo.xxxxxxxx | June 5, 2026

Independent research draft. Not an enacted law and not legal advice; not endorsed by the FDA, HHS, or any Member of Congress.

Subject

Opening statement

Chairman, Ranking Member, and Members of the Subcommittee: thank you for the opportunity to submit this statement. It describes a narrow, growing safety gap in how AI now writes the code that moves robots in cancer trials, and a low-cost rule that would close it.

Three points for the record

1. Embodiment changes the stakes: a verification done after code runs is, for a moving instrument, done too late.
2. The gate is feasible today; it has been run, reproducibly, on a surgical-humanoid control code-base.
3. The rule is additive, building on the section 515C change-control authority the FDA already uses.

Figure placeholder (Images/testimony-evidence.png). The float, caption, and \label mirror the VVUQ-02 figures; replace the box with \includegraphics.

Figure 1: Figure 1. Placeholder schematic of the evidence offered for the record.

Anticipated questions

- Q.** Is this technically possible before code runs?
- A.** Yes; demonstrated at 51/51 and 172/172 tests with the verification fraction held at 1.0.

Q. Why amend the FD&C Act rather than write a new law?

A. The substantive content already lives in device, privacy, and civil-rights authorities; one sequencing rule is what is missing.

Research-influence appendix

Influence	Where discussed	Status
State human-in-the-loop laws	savings-clause design	enacted, various States
FDA PCCP guidance	gate framing	final guidance
ASME V&V 40-2018	credibility criteria	consensus standard

Table 1: Table 1. Emerging influences, kept out of the operative text (modeled on VVUQ-05 Table 1).

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Cite This Brief

Kawchak K. Testimony and Research-Influence Brief for H. R. 9510. Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

Data Availability

Supporting records are openly available in the [kevinkawchak/cancer-automated](#) and [kevinkawchak/physical-ai-oncology-trials](#) GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

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